

**B.Tech. Degree V Semester (Supplementary) Examination,
July 2009**

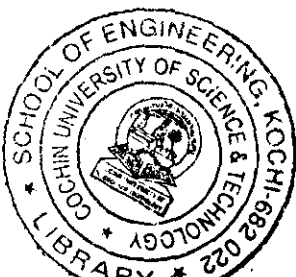
**ME 505 HEAT AND MASS TRANSFER
(2002 Scheme)**

Time: 3 Hours

Maximum Marks: 100

- I a) Define Fourier law of heat conduction. Give the units and physical significance of each term appearing in the Fourier equation. (8)
- b) A furnace wall is made up of steel plate 10mm thick ($k = 62.8 \text{ kJ/m-hr-deg}$) lined on inside with silica bricks 150mm thick ($k = 7.32 \text{ kJ/m-hr-deg}$) and on the outside with magnesia bricks 200mm thick ($k = 18.84 \text{ kJ/m-hr-deg}$). The inside and outside surface of the wall are at temperatures of 650°C and 125°C respectively. Make calculations for the heat loss from the unit area of the wall. (12)
- OR**
- II a) Explain the following terms: (8)
- i) Conduction shape factor
- ii) Critical insulation thickness.
- b) A rod of 10mm square section and 160mm length with thermal conductivity of 50 W/m-deg protrudes from a furnace wall at 200°C , and is exposed to air at 30° with convection coefficient $20 \text{ W/m}^2 - \text{deg}$. Make calculations for the heat convected upto 80mm and 158mm lengths and comment on the result. Adopt a long fin model for the arrangement. (12)
- III a) Differentiate between the mechanisms of heat transfer by free and forced convection. Mention some of the areas where these mechanisms are predominant. (8)
- b) A heat-treat steel plate measures $3\text{m} \times 1\text{m}$ and is initially at 30°C . It is cooled by blowing air parallel to 1m edge at 9 km/hr . If the air is at 10°C , calculate the convective heat transfer from both sides of the plate. (12)
- OR**
- IV a) How does film wise condensation differ from drop wise condensation? Which type has a higher heat transfer film co-efficient and point out the reason for it? (8)
- b) A steam pipe 50mm diameter and 2.5m long has been placed horizontally and exposed to still air at 25°C . If the pipe wall temperature is 295°C , determine the rate of heat loss. At the mean temperature of 160°C , the thermo-physical properties of air are: $k = 0.0364 \text{ W/m-deg}$, $\nu = 30.09 \times 10^{-6} \text{ m}^2/\text{s}$, $\text{Pr} = 0.682$, $\beta = 2.31 \times 10^{-3} \text{ per K}$. (12)
- V a) Explain the terms absorptivity, reflectivity and transmissivity of radiant energy. How they are related to each other for a black body and an opaque body. (8)
- b) A furnace emits radiation at 2000K . Treating it as a black body radiation, calculate the
- i) Monochromatic radiant flux density at 1μ wavelength.
- ii) Wavelength at which emission is maximum and the corresponding radiating flux density.
- iii) Total emissive power. (12)
- OR**
- VI a) Define the radiation shape factor and give its salient features. (8)
- b) Determine the radiating heat flux between two closely spaced, black parallel plates radiating only to each other if their temperatures are 850K and 425K respectively. Recalculate the heat flux presuming that each of the parallel plates has an emissivity of 0.5. In each case, the plates have an area of 4m^2 . (12)

(Turn over)



- VII a) What is a heat exchanger? How heat exchangers are classified? Sketch a shell and tube heat exchanger. (8)
- b) Exhaust gases ($c_p = 1.12 \text{ kJ/kg-deg}$) flowing through a tubular heat exchanger at the rate of 1200 kg/hr are cooled from 400°C to 120°C . The cooling is affected by water ($c_p = 4.18 \text{ kJ/kg-deg}$) that enters the system at 10°C at the rate of 1500 kg/hr. If the overall heat transfer co-efficient is $500 \text{ kJ/m}^2\text{-hr-deg}$, what heat exchanger area is required to handle the load for
- Parallel flow
 - Counter flow arrangement. (12)
- OR**
- VIII a) Derive an expression logarithmic mean temperature difference in the case of a counter flow heat exchanger. (8)
- b) Hot water having specific heat 4200 kJ/kg K flows through a heat exchanger at the rate of 4 kg/min. with an inlet temperature of 100°C . A cold fluid having specific heat of 2400 kJ/kg K flows in at a rate of 8 kg/min. with inlet temperature of 20°C . Make calculations for the maximum possible effectiveness if the fluid flow confirms to (i) parallel flow arrangement (ii) counter flow arrangement. (12)
- IX a) State and explain different modes of mass transfer. (8)
- b) A plastic membrane 0.25 mm thick has hydrogen gas maintained at pressure of 2.5 bar and 1 bar on its opposite sides. The binary diffusion co-efficient of hydrogen in the plastic is $8.5 \times 10^{-8} \text{ m}^2/\text{s}$ and the solubility of hydrogen in the membrane is $1.5 \times 10^{-3} \text{ kg-mol/m}^2\text{-bar}$. Under uniform temperature conditions of 25°C , determine
- molar concentrations of hydrogen at opposite faces of the membrane
 - molar and mass diffusion flux of hydrogen through the membrane. (12)
- OR**
- X a) Explain Ficks law of diffusion and what are its limitations? (8)
- b) Water is available at the bottom of a well, which is 2.5 m in diameter and 5 m in deep. Estimate its diffusion rate into dry atmospheric air at 25°C . The diffusion co-efficient for the system is approximated to be $0.0925 \text{ m}^2/\text{hr}$ and the atmospheric pressure is 1.032 bar. (12)
